LyMAPPO: Lyapunov-Constrained Independent Learning for Ultra-High Reliability in Dense Multi-Band Wi-Fi

Jiha Kim, Seunghyun Park, and Hyunhee Park

***Abstract*—IEEE 802.11bn defines ultra-high reliability (UHR) through per-BSS deadline-violation-rate targets. We study these constraints in dense multi-band deployments with heterogeneous AP link sets. LyMAPPO combines independent per-AP PPO with drop-inclusive rate-form Lyapunov duals, one dual-derived congestion price per beacon for cross-BSS coordination (PX), and an auxiliary lower-tail quantile critic (QR). Training and evaluation use an ns-3-in-the-loop environment with 16 APs, three bands, shared-buffer multi-link operation, and Markov channels. Across 16 seeds with 110 s episodes, LyMAPPO achieves 15.1/16 feasible BSSs and a network p_{99} of 0.0017, which is 6.1–$25\times$ lower than for three static baselines. Feasibility and network-p_{99} differences remain significant after Holm correction, whereas deep-tail comparisons are weaker. LyMAPPO yields the best point estimates against MAPPO and PPO-Lagrangian baselines trained in the same environment, although the differences are not significant with eight seeds. It retains its advantage over static baselines across a 40–80 pkt/s/STA load sweep. Federated aggregation matches independent learning on the training distribution but reduces held-out reliability, particularly for APs with limited spectrum access. The drift-plus-penalty analysis accounts for clipping and establishes a run-time dual-stability condition under the Slater assumption.**

***Index Terms*—Wireless LAN, multi-link operation, multi-agent reinforcement learning, Lyapunov optimization, ultra-high reliability.**

I. Introduction

IEEE 802.11bn ("Wi-Fi 8") includes *ultra-high reliability* (UHR) objectives in addition to peak-throughput targets. Representative per-BSS requirements limit deadline-violation rates to 10^{-2} at the p_{99} scale and 10^{-3} at the $p_{99.9}$ scale for interactive and industrial traffic [1]. We consider dense enterprise deployments in which all APs are within mutual carrier-sense range, multi-link operation (MLO) uses the 2.4/5/6 GHz bands, and APs have heterogeneous link sets. Under these conditions, the band-sharing graph determines contention among BSSs.

At the 16-AP operating point considered in this study, balanced round-robin satisfies the UHR constraints for an average of 9.0 BSSs, RSSI-based selection for 7.8, and the published least-congested-interface rule (SLCI) for 3.9 (Section V). SLCI causes multiple BSSs to select the same temporarily underutilized band, thereby increasing contention. These results indicate that per-BSS violation-rate constraints require adaptation to traffic load, channel state, and the policies of neighboring BSSs.

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Prior learning-based WLAN controllers have primarily optimized throughput or mean delay using DRL for MLO flow allocation [2], [3] or bandits for spatial reuse [4]. Lyapunov-constrained RL has been applied to violation-probability constraints through drift-plus-penalty duals [5], [6], including centralized cellular scheduling [7], but not decentralized Wi-Fi control. Federated learning has also been proposed for multi-AP Wi-Fi [8], [9]; however, reported gains over independent learners decrease as the number of BSSs increases [9]. This study addresses two questions: how to enforce per-AP deadline-violation-rate constraints in dense multi-band Wi-Fi, and how the choice between parameter sharing and low-rate congestion signaling affects reliability under heterogeneous link availability.

LyMAPPO trains one PPO policy per AP using rate-form Lyapunov duals. The virtual-queue increments are per-slot violation rates over all decided packets, including both deliveries and drops, which avoids survivor bias and bounds each increment (Section IV). For cross-BSS coordination, each AP broadcasts the sum of its dual variables as a one-scalar congestion price (**PX**). A neighboring AP incorporates this price according to its use of the corresponding band. PX is a heuristic proxy for the cross-BSS term in the network Lagrangian and does not share or aggregate policy parameters. Its isolated effect is a statistically significant reduction in inter-seed dispersion rather than a significant change in the mean (Section V-E). The auxiliary **QR** component replaces the mean critic baseline with a lower-tail CVaR quantile baseline. Its isolated effect is not statistically significant, and we therefore treat QR as an auxiliary component. Training and evaluation use the same ns-3-in-the-loop contention model with 802.11ax per-link PHY/MAC devices, shared-buffer MLO, and Markov channels.

We compare PX with parameter aggregation using FedAvg [10], q-FFL [11], AFL [12], and clustered FL [13], all based on the same underlying learner. The federated and independent variants have comparable performance on the training distribution (Fig. 5). On held-out channel conditions, however, all evaluated federated variants have lower reliability than the independent base learner, with most violations occurring in the class with the least spectrum access (Section V-C). PX communicates one scalar per beacon while retaining separate per-AP policies.

This paper makes four contributions. **(C1)** We formulate LyMAPPO, a rate-form Lyapunov-constrained independent MARL method with one-scalar cross-BSS congestion signaling (PX) and an auxiliary lower-tail quantile critic (QR). The analysis includes the clipping deficit and a run-time dual-stability condition for Slater-feasible APs (Section IV-A).

(C2) We implement an ns-3-in-the-loop training system with shared-buffer MLO, drop-inclusive accounting, and a pre-specified evaluation protocol (Section V-A). **(C3)** At 16-BSS density, LyMAPPO achieves 15.1/16 feasible BSSs and a network p_{99} that is 6.1–25$\times$ lower than the static baselines. Feasibility and network-p_{99} differences remain significant after Holm correction. LyMAPPO also has the best point estimates among the learned baselines trained in the same environment, although these differences are not significant with eight seeds (Section V-B). **(C4)** We evaluate four federated aggregation methods and relate their held-out reliability to the loss of per-AP policy specialization under parameter averaging (Section V-C).

II. Related Work

a) MLO traffic allocation: Multi-link operation is a principal latency mechanism in Wi-Fi 7/8 [1]. Measurement studies report order-of-magnitude latency reductions under symmetric link occupancy, but performance can degrade under asymmetric occupancy [14]. Published allocation methods include congestion-aware flow placement (SLCI/MCAA) [15], adaptive scoring and minimum-queue steering evaluated with ns-3 and ns3-ai [16], and a latency-optimal static split for MLD/SLD coexistence [17]. In our 16-BSS environment, concurrent least-congested-interface decisions increase concentration on the same band (Section V-B). The 802.11be MLO delay behavior in ns-3 is evaluated in [18].

b) Learning-based WLAN control: DRL has been applied to MLO flow allocation (multi-headed recurrent SAC, +35.2% throughput-drop ratio vs. SLCI [2]) and cooperative channel selection [3], and decentralized bandits to spatial reuse [4]. These works optimize throughput or mean delay; none enforces a deadline-violation-*rate* constraint, and none reports per-BSS reliability feasibility.

c) Federated RL for Wi-Fi: Federated DRL improves intra-BSS channel access under a single AP [8]. For multi-AP MLO link activation, federated max-min MAB improves performance at low density, while its advantage over independent learners decreases by 16 APs [9]. We evaluate FedAvg [10], fairness-weighted q-FFL [11] and AFL [12], and clustered FL [13] at 16-BSS density with heterogeneous link sets. The results separate training-distribution performance from held-out reliability and quantify the reduction in per-AP policy specialization after aggregation (Section V-C). PX provides cross-BSS congestion information without aggregating the actor parameters (Section IV).

d) Lyapunov-constrained RL: Drift-plus-penalty optimization [5], [19] hybridized with DRL is established for queue-stable offloading [6], and Lyapunov-guided MARL enforces an explicit 1% delay-violation probability in centralized cellular scheduling [7]. To our knowledge no prior work brings this constraint class to per-AP Wi-Fi MLO control with survivor-bias-free rate-form duals, nor couples the duals across BSSs as congestion prices.

III. System Model and Problem Formulation

a) Deployment: We consider an enterprise WLAN with N=16 APs arranged on a 4×4 grid with 8 m spacing. All APs are within mutual carrier-sense range. Each AP serves $|\mathcal{S}_i|$=5 stations placed on a 3 m ring, for a total of 80 stations (Fig. 1). Stations associate only with their designated AP using a per-BSS SSID; inter-BSS interactions therefore arise from OBSS contention on shared bands.

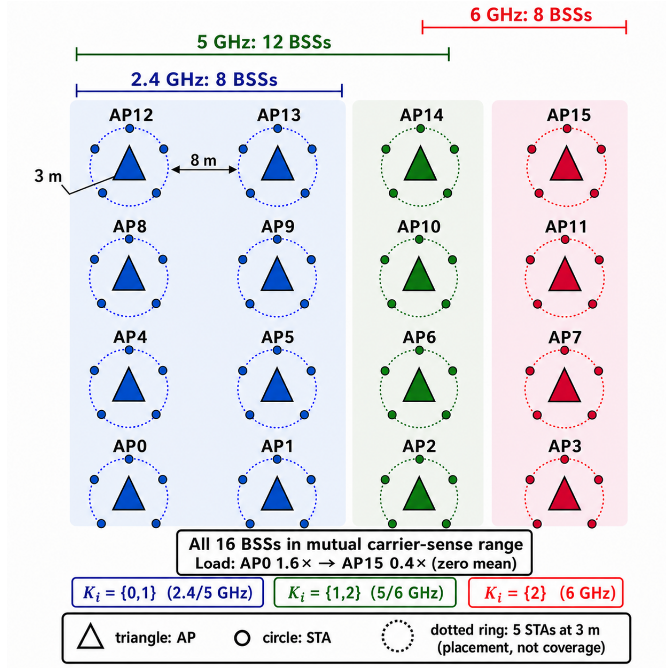

Fig. 1. Network topology with 16 APs on a 4×4 grid at 8 m spacing, all within mutual carrier-sense range. Each AP serves five STAs placed on a 3 m ring; the dotted rings indicate placement rather than RF coverage. Asymmetric link sets $\mathcal{K}_i$ are assigned by column. The band brackets show the three OBSS contention domains: 8 BSSs at 2.4 GHz, 12 at 5 GHz, and 8 at 6 GHz. The corresponding bandwidths are 20, 40, and 40 MHz. The zero-mean offered load decreases from 1.6$\times$ at AP 0 to 0.4$\times$ at AP 15.

b) Heterogeneous multi-band links: We model three links with different bandwidths: link 0 at 2.4 GHz/20 MHz, link 1 at 5 GHz/40 MHz, and link 2 at 6 GHz/40 MHz. The link set $\mathcal{K}_i$ is assigned according to $i \bmod 4$ as $\{0,1\},\{0,1\},\{1,2\},\{2\}$. Thus, the band-sharing graph determines contention among BSSs. APs with $\mathcal{K}$=$\{2\}$ have no link diversity, whereas those with $\mathcal{K}$=$\{0,1\}$ have no 6 GHz access. Section V-B identifies the class for which these restrictions limit reliability at the selected operating point.

c) Channel dynamics: For each station–link pair, large-scale channel quality follows a three-state Markov chain with $h\in\{0.9,0.7,0.5\}$. These states correspond to excess losses of $2/6/10$ dB applied symmetrically to all AP paths toward the station. Transitions occur between adjacent states with a mean dwell time of 200 ms. This model satisfies the finite-state, bounded-mixing assumption used in the analysis and permits closed-loop rate control to track the approximately 4 dB state changes. In a calibration variant with states $\{0.9,0.5,0.3\}$, the larger transitions caused persistent rate-control retries and eliminated Slater feasibility; that variant is not used in the evaluation.

d) Traffic and load heterogeneity: Each station generates constant-bit-rate UHR traffic at λ=60 pkt/s with 1500 B packets. A zero-mean linear load spread varies the per-AP offered load from $\times 1.6$ at AP 0 to $\times 0.4$ at AP 15 while preserving the network mean. This construction produces different constraint loads across BSSs.

e) Shared-buffer MLO queueing (service-time routing): Each station maintains a single upper-layer buffer. A head-of-line packet enters link k's MAC queue only when that queue is below the target depth of eight packets. Link selection is therefore applied at service time, and packets remaining in the shared buffer can be reassigned after a policy change. Per-link EDCA MAC queues implement contention. Packets older than $\max(5L_{99.9}, 100\,\mathrm{ms})$ are discarded from either queueing stage and counted as decided violations. This drop-inclusive accounting prevents survivor bias.

f) PHY/MAC: The environment is ns-3.40 with 802.11ax PHY/MAC (MLO emulated by per-link single-link devices), ideal rate adaptation, RTS/CTS off; the learner interacts in-the-loop through shared memory at a 20 ms macro-slot (Section V-A).

A. Problem: CMDP with per-AP UHR constraints

For percentile $\ell \in \{99, 99.9\}$, let $v_i^\ell(t)$ denote the number of packets associated with AP i whose latency exceeds deadline L_ℓ in macro-slot t. Let $d_i^\ell(t)$ denote the number of packets decided in that slot, including both delivered and aged-out packets. The per-AP UHR requirement is the following long-run violation-rate constraint:

$$\bar{V}_i^\ell = \limsup_{T\to\infty} \frac{1}{T} \sum_{t<T} \frac{v_i^\ell(t)}{d_i^\ell(t)} \;\le\; \varepsilon_\ell, \qquad \forall i, \ell, \tag{1}$$

with $\varepsilon_{99} = 10^{-2}$, $\varepsilon_{99.9} = 10^{-3}$. Deadlines are density-scaled: canonical single-BSS targets are $(5, 10)$ ms; for the 16-AP dense multi-band deployment we use $(L_{99}, L_{99.9}) = (20, 40)$ ms, since the structural access-latency floor of a band-sharing CSMA collision domain grows with the number of contending BSSs and fixed small deadlines would void Slater feasibility for *every* policy.

Remark 1 (Per-slot vs. packet-average rate)**.** *Equation* (1) *is the Cesàro average of the per-slot rates used in the dual update of Section IV. The empirical results use the packet-average ratio* $\sum_t v / \sum_t d$. *We logged both quantities per AP in one held-out diagnostic run to evaluate their agreement. For nonempty slots, the decided count was in* $[d_{\min}, d_{\max}]{=}[1, 30]$*; a minority of slots at lightly loaded APs were empty and used the* $g{=}-\varepsilon_\ell$ *convention in Section IV. Violations did not concentrate in slots with small d, and the two rates differed by less than one third at every AP. For example, the per-slot and packet-average* p_{99} *values at the highest-load AP were* 0.0022 *and* 0.0027*, respectively. In the same run, the per-slot rate satisfied both targets at all APs: the maximum values were* $\bar{V}^{99}{=}0.0022{<}10^{-2}$ *and* $\bar{V}^{99.9}{=}0.0006{<}10^{-3}$*. This diagnostic does not establish a per-seed guarantee, but it provides a direct check that the reported packet-average metric is consistent with the constrained per-slot quantity.*

The CMDP maximizes throughput utility subject to (1):

$$\max_{\pi} \; \limsup_{T\to\infty} \frac{1}{T} \mathbb{E}_\pi\Big[\sum_{t<T}\sum_i u_i(t)\Big] \quad \text{s.t.} \quad \bar{V}_i^\ell \le \varepsilon_\ell \;\; \forall i, \ell, \tag{2}$$

where $u_i(t)$ is AP i's delivered packets in slot t (normalized per station); dropped packets earn nothing.

IV. LyMAPPO: Lyapunov-Constrained Per-AP Learning

Figure 2 summarizes the architecture and the separation between training and decentralized execution. The components are defined below.

a) Rate-form virtual queues: Let $g_i^\ell(t) := v_i^\ell(t)/d_i^\ell(t) - \varepsilon_\ell$ be the per-slot violation-rate slack (d drop-inclusive; a slot with no decided packets contributes $g = -\varepsilon_\ell$ by convention). Each AP maintains one virtual queue (dual variable) per constraint,

$$Z_i^\ell(t{+}1) = \mathrm{clip}_{[0, Z_{\max}]}\big(Z_i^\ell(t) + g_i^\ell(t)\big), \qquad Z_{\max} = 10, \tag{3}$$

The increments satisfy $|g| \le 1$ for all traffic loads. By contrast, the magnitude of the count-form term $v - \varepsilon d$ scales with traffic and can destabilize the reward under saturation. Slots with $g < 0$ reduce the dual variable.

b) Reward: The drift-plus-penalty per-slot reward is

$$r_i(t) = V\, u_i(t) - \textstyle\sum_\ell Z_i^\ell(t)\, g_i^\ell(t), \tag{4}$$

with the *pre-update* dual multiplying the slack that updates it (Section IV-A); $V{=}1$ throughout.

c) State and action: For each station and link, the local state includes ns-3-measured CSI, shared-buffer and MAC backlog ($\log(1{+}\cdot)$), head-of-line age normalized by $L_{99.9}$, channel-busy ratio, the two dual variables normalized by $Z_{\max}$, and link mask $\mathbf{1}_{\mathcal{K}_i}$. The action selects one link per station from mask-constrained logits, with the same mask used during training and evaluation. The interface includes a multi-AP coordination mode, but it is fixed to 'none' because Co-TDMA serializes the dense medium by a factor of N and Co-OFDMA is outside the PHY model. Link routing is therefore the only learned control variable.

d) Learning: Each AP trains a separate PPO actor using (4) and GAE advantages. A centralized critic has N output heads, with head i estimating AP i's return from the joint observation. This design provides centralized training and decentralized execution (CTDE) while preserving AP-specific advantages under heterogeneous rewards. Formally, decentralized execution defines a constrained Dec-POMDP, and the per-AP CMDP in Section III-A is the local relaxation monitored by the dual variables. Actor parameters are not shared or aggregated across APs. Section V-C evaluates how parameter aggregation affects held-out reliability under heterogeneous link sets. Model selection uses the per-AP CMDP infeasibility score $\frac{1}{N}\sum_i \sum_\ell [\hat{V}_i^\ell - \varepsilon_\ell]^+/\varepsilon_\ell$ rather than a network mean that could average over infeasible APs.

e) Cross-BSS price coupling (PX): Independent actors do not directly observe the constraint state of neighboring APs, although their link choices affect shared-band contention. This limitation contributes to seed-dependent variation in which $\{2.4, 5\}$-class APs remain feasible (Section V-B). PX introduces a congestion-price proxy based on the neighboring APs' dual sums. This proxy is heuristic rather than a first-order approximation because it does not estimate $\partial g_j / \partial\, \mathrm{use}_{i,k}$. Each

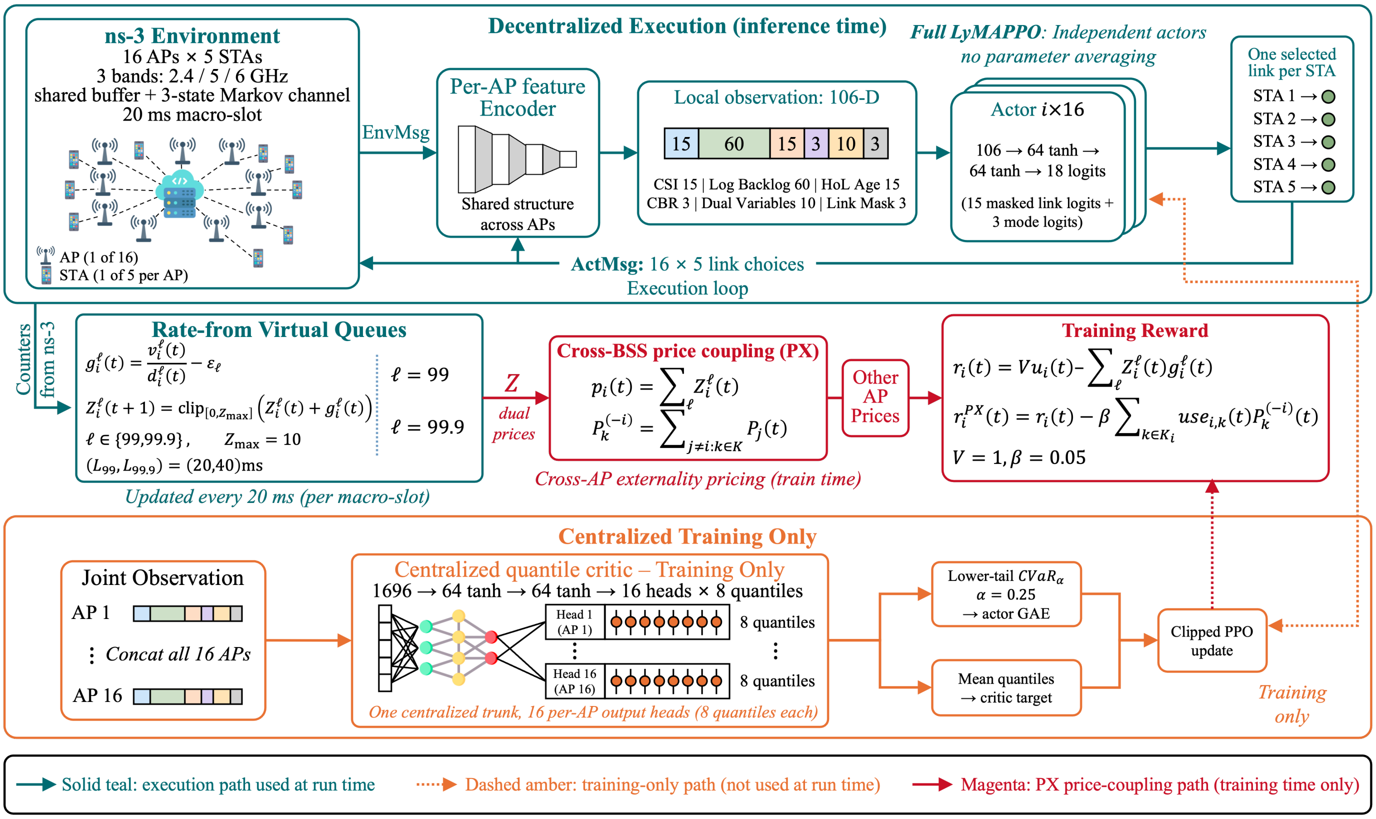

Fig. 2. LyMAPPO model and training loop. During decentralized execution (top, teal), ns-3 provides per-slot counters through EnvMsg. Each AP encodes a 106-D local observation containing CSI, log queue length, HoL age, CBR, dual variables, and a link mask. Its independent actor (106→64→64→18 with masked logits) selects one link per STA and returns the selections through ActMsg. Rate-form duals are updated every 20 ms and define the PX prices p_i and the band aggregate P_k^{-i} used in the training reward. During centralized training (bottom, amber), the quantile critic (1696→64→64→16 heads × 8 quantiles) provides a lower-tail CVaR baseline for clipped-PPO updates. The amber and magenta paths are inactive during inference.

AP broadcasts the scalar $p_i = \sum_\ell Z_i^\ell$, which can be carried in a beacon information element. The price of band k for AP i is $P_k^{-i} = \sum_{j\neq i:\, k\in\mathcal{K}_j} p_j$, and the training reward is

$$r_i^{\mathrm{PX}}(t) = r_i(t) - \beta \sum_{k\in\mathcal{K}_i} \mathrm{use}_{i,k}(t)\, P_k^{-i}(t), \qquad \beta = 0.05, \quad (5)$$

where $\mathrm{use}_{i,k}$ is the fraction of AP i's stations routed to band k in the slot. PX affects training only and does not modify decentralized execution. Equation (5) reduces to (4) when $\beta \to 0$ or $Z \to 0$. Relative to the analysis, PX is an $O(\beta)$ perturbation of the exact surrogate and is included in the C_{RL} slack in Theorem 1. The training implementation exchanges prices without loss every 20 ms. A deployment could include an 8-bit quantization of p_i in beacons at an approximately 100 ms interval. Beacon losses and stale price information are not modeled in this study.

f) Distributional tail critic (QR): A mean critic may provide limited resolution for the rare slots that receive large constraint penalties. Each critic head therefore estimates $K{=}8$ return quantiles using pinball loss. The actor advantage uses the mean of the lower-tail quantiles (CVaR_α with $\alpha{=}0.25$), whereas the critic regression target retains the mean baseline (double-GAE). This construction gives greater weight to lower-return slots without changing the mean value target. QR is treated as an auxiliary component because its isolated effect on the pre-specified metrics is not statistically significant (Section V-E).

The full method combines (3), (5), and QR. The ablation results in Section V-E show lower across-seed dispersion for this configuration. Table I reports the results of the full method.

A. Analysis

Assumption 1. *(i) Bounded decided counts: $d_i^\ell(t) = 0$ or $d_i^\ell(t) \in [d_{\min}, d_{\max}]$, $d_{\min} > 0$, with $g_i^\ell(t) = -\varepsilon_\ell$ when $d_i^\ell(t) = 0$; (ii) finite-state Markov channel with bounded mixing; (iii) Slater: some policy attains $\bar{V}_i^\ell \le \varepsilon_\ell - \delta$, $\delta > 0$, for all i, ℓ.*

Proposition 1 (Exact rate-form DPP surrogate)**.** *Let $\Phi(t) = \frac{1}{2}\sum_{i,\ell}(Z_i^\ell(t))^2$. Ignoring the upper clip, $\Phi(t{+}1) - \Phi(t) \le \sum_{i,\ell} Z_i^\ell g_i^\ell + \frac{1}{2}\sum_{i,\ell}(g_i^\ell)^2$ with $\frac{1}{2}\sum_{i,\ell}\mathbb{E}[(g_i^\ell)^2] \le B := NK/2$ for K constraints per AP (here $K{=}2$), since $|g| \le 1$. Hence maximizing $\sum_i r_i$ in (4) minimizes exactly the policy-dependent part of the drift-plus-penalty bound $\Delta(t) - V\,\mathbb{E}[u(t) \mid Z]$.*

Proof. The unclipped update is $Z_i^\ell(t{+}1) = [Z_i^\ell(t) + g_i^\ell(t)]^+$. Since $([x]^+)^2 \le x^2$,

$$\tfrac{1}{2}\left(Z_i^\ell(t{+}1)\right)^2 - \tfrac{1}{2}\left(Z_i^\ell(t)\right)^2 \le Z_i^\ell(t)\, g_i^\ell(t) + \tfrac{1}{2}\left(g_i^\ell(t)\right)^2,$$

and summing over the NK pairs (i, ℓ) gives the stated drift inequality. In a nonempty slot, $g_i^\ell = v_i^\ell / d_i^\ell - \varepsilon_\ell$ with

$0 \le v_i^\ell \le d_i^\ell$ and $d_i^\ell \ge 1$, where the decided count includes drops. When $d_i^\ell = 0$, the convention gives $g_i^\ell = -\varepsilon_\ell$. Thus, $g_i^\ell \in [-\varepsilon_\ell, 1-\varepsilon_\ell] \subset [-1,1]$ deterministically and $\frac{1}{2}\sum_{i,\ell}\mathbb{E}[(g_i^\ell)^2] \le NK/2 =: B$. This bound does not require an arrival-moment assumption. Taking the conditional expectation given $Z(t)$ yields

$$\Delta(t) - V\mathbb{E}[u(t) \mid Z] \le B - \mathbb{E}\Big[\underbrace{Vu(t) - \textstyle\sum_{i,\ell} Z_i^\ell(t) g_i^\ell(t)}_{=\ r(t)\ \text{by (4)}}\Big| Z\Big],$$

where the identification with (4) holds because the implementation forms the reward with the *pre-update* duals. Maximizing $\mathbb{E}[r(t) \mid Z]$ therefore minimizes the policy-dependent part of the bound exactly. (In the count form $g = v - \varepsilon d$ the offset εd is itself policy-dependent here, since decided counts depend on drain and discard decisions; the rate form removes this slack.) □

Theorem 1 (Constraint satisfaction)**.** *Under Assumption 1, if the learned policy is within $C_{\mathrm{RL}} \ge 0$ of the per-slot minimizer of the bound in Proposition 1, then with $Z(0) = 0$, telescoping yields*

$$\frac{1}{T}\sum_{t<T}\mathbb{E}[u(t)] \;\ge\; U^\star - \frac{B + C_{\mathrm{RL}}}{V}, \tag{6}$$

$$\frac{1}{T}\sum_{t<T}\sum_{i,\ell}\mathbb{E}\big[Z_i^\ell(t)\big] \;\le\; \frac{B + C_{\mathrm{RL}} + V(U_{\max} - U^\star)}{\delta},$$

and mean-rate stability of the unclipped queues gives $\limsup_T \frac{1}{T}\sum_{t<T}\big(v_i^\ell/d_i^\ell - \varepsilon_\ell\big) \le 0$, which is (1)*. With clipping, the bound becomes $\bar{V}_i^\ell \le \varepsilon_\ell + \bar{\rho}_i^\ell$, where $\rho_i^\ell(t) = [Z_i^\ell(t) + g_i^\ell(t) - Z_{\max}]^+$ is the clipping deficit. Selecting $Z_{\max}$ above the equilibrium backlog, which is $O(V/\delta)$, limits $\bar{\rho}$. In the experiments, the dual variables peak near 5 during the transient and converge to approximately* 0.4*, below $Z_{\max}$=10.*

Proof. Let $\pi^\star$ be the Slater policy of Assumption 1(iii), attaining $\mathbb{E}[g_i^\ell \mid Z] \le -\delta$ for all (i,ℓ) with utility $U^\star$. A policy within C_{RL} of the per-slot minimizer of the bound in Proposition 1 does, in particular, at most C_{RL} worse than $\pi^\star$:

$$\Delta(t) - V\mathbb{E}[u(t) \mid Z] \le B + C_{\mathrm{RL}} - VU^\star - \delta\sum_{i,\ell} Z_i^\ell(t).$$

Summing over $t = 0, \ldots, T-1$ with $\Phi(0) = 0$ and $\Phi(T) \ge 0$,

$$\begin{aligned} 0 \le \mathbb{E}[\Phi(T)] &\le T(B + C_{\mathrm{RL}}) - VTU^\star \\ &\quad + V\sum_{t<T}\mathbb{E}[u(t)] - \delta\sum_{t<T}\sum_{i,\ell}\mathbb{E}[Z_i^\ell(t)]. \end{aligned}$$

Dropping the nonnegative δ-term and dividing by VT gives the utility bound in (6); bounding $\mathbb{E}[u(t)] \le U_{\max}$ instead and dividing by δT gives the backlog bound. For constraint satisfaction, the unclipped update satisfies $Z_i^\ell(t{+}1) \ge Z_i^\ell(t) + g_i^\ell(t)$, hence $Z_i^\ell(T) \ge \sum_{t<T} g_i^\ell(t)$; the same telescoping gives $\mathbb{E}[\Phi(T)] = O(VT)$, so by Jensen $\mathbb{E}[Z_i^\ell(T)] \le \sqrt{2\,\mathbb{E}[\Phi(T)]} = O(\sqrt{VT})$ and

$$\limsup_{T\to\infty}\frac{1}{T}\sum_{t<T}\mathbb{E}\big[v_i^\ell/d_i^\ell - \varepsilon_\ell\big] \le \limsup_{T\to\infty}\frac{\mathbb{E}[Z_i^\ell(T)]}{T} = 0,$$

which is (1). With the clip $Z(t{+}1) = \min(Z_{\max}, [Z(t) + g(t)]^+)$, each saturation event discards exactly $\rho_i^\ell(t) = [Z_i^\ell(t) + g_i^\ell(t) - Z_{\max}]^+$ from the accumulated sum, so $Z_i^\ell(T) \ge \sum_{t<T} g_i^\ell(t) - \sum_{t<T}\rho_i^\ell(t)$ and the same argument yields $\bar{V}_i^\ell \le \varepsilon_\ell + \bar{\rho}_i^\ell$. Finally, since the post-clip queue obeys $Z_i^\ell(t) \le Z_{\max}$, we have $\rho_i^\ell(t) \le g_i^\ell(t) \le 1$ per event, and the backlog bound gives a stationary mean $\bar{Z} = O(V/\delta)$; Markov's inequality then bounds the saturation frequency, hence $\bar{\rho}_i^\ell$, by $O\big((V/\delta)/Z_{\max}\big)$. Choosing $Z_{\max} = \omega(V/\delta)$ drives the deficit to zero; at the operating point the duals equilibrate at $\approx 0.4 \ll Z_{\max}$=10 after a transient peak ≈ 5 (Remark 2), making saturation, and with it $\bar{\rho}$, negligible. □

Remark 2 (Guarantee vs. metric)**.** *Theorem 1 bounds the long-run average violation rate rather than a per-sample quantile. We therefore report mean±std across seeds and seed-level statistical tests in addition to feasibility counts.*

Remark 3 (Empirical consistency of the constants)**.** *The constants can be compared with 960 logged training iterations for each of the full and base learners.* (i) Increments: *the largest per-rollout change in the across-AP mean of Z^{99} is* 0.33*, close to the rate-form drain limit $32\,\varepsilon_{99}$=0.32 for a 32-slot rollout. A count-form dual would instead scale with traffic.* (ii) Clipping deficit: *after a transient peak near 5, $\sum_\ell \bar{Z}^\ell$ converges to approximately* 0.34–0.38*. A conservative maximum-AP bound, computed as* $\mathrm{mean}{+}\sqrt{15}\,\mathrm{std}$ *over 16 APs, identifies only* 0.8–1.3% *of iterations as potentially reaching $Z_{\max}$=10, all during the transient. No clipping deficit is observed after convergence.* (iii) Scope of the Slater assumption: *$p_{99.9}$ is the binding constraint, with network-level slack of approximately* 10^{-4} *compared with* $8.6{\times}10^{-3}$ *for p_{99}. Assumption 1(iii) applies per AP. Residual infeasibility occurs in the $\{2.4, 5\}$ class for which $\delta \le 0$ at the feasibility boundary (Section V-B). Theorem 1 therefore establishes virtual-queue stability only for Slater-feasible APs and does not establish feasibility for that class. The constant C_{RL} is an unmeasured a-priori policy optimality gap; dual stability and the clipping deficit, but not C_{RL}, can be evaluated at run time.*

V. Evaluation

A. Experimental setup

a) In-the-loop system: The implementation uses ns-3.40 and ns3-ai shared memory, with one EnvMsg/ActMsg exchange per 20 ms macro-slot. Training consists of 100 iterations with 32-slot rollouts in a persistent episode. Evaluation loads the selected checkpoint and runs 110 s episodes after excluding a 1 s settling interval. Each AP contributes a median of 33,000 decided packets per run and at least 10^5 packets per training seed. The held-out channel seeds are $\{10, 12, 18\}$. Training uses $s \in \{0..7\}$ for all arms (n=8), extended to $s \in \{0..15\}$ for the full and base methods (n=16). For learned methods, the statistical unit is the training seed and each observation is averaged over the three held-out channel seeds. Static baselines have no training seed and are evaluated on eight channel seeds $\{10, 12, 13, \ldots, 18\}$. Channel seed 11 causes a pre-existing ns-3 PHY assertion at phy-entity.cc:509 and was excluded and replaced; four other evaluation cells that encountered the same assertion were also excluded. The load sweep retrains the full method at $\lambda \in \{40, 80\}$ using $s \in \{0..7\}$; one training run at λ=80 did not complete and was excluded. Static baselines use three held-out seeds per load. The sensitivity and load-sweep studies in Table IV and Fig. 6 use an 11 s development protocol. For configurations evaluated with both durations, the feasibility estimates differ by at most 0.4 APs. All exclusions are documented and are consistent across episode durations.

b) Baselines and arms: The non-learned baselines are balanced round-robin (RR), RSSI-maximum link selection, and the published least-congested-interface policy (SLCI) [15]. The federated variants are FedAvg [10], q-FFL [11], AFL [12], and link-set-clustered FedAvg [13] (Section V-C). Each federated method aggregates the base learner without PX or QR. The ablations remove the dual layer (no-Z) or individual PX/QR components (Section V-E). All methods use the same environment and evaluation protocol.

c) Implementation details: Each actor is a two-layer MLP with a 106-dimensional local state, two 64-unit tanh layers, and 18 output logits; unused mode logits are fixed. The centralized critic maps the 1696-dimensional joint observation through two 64-unit tanh layers to 16 heads with K=8 quantiles each. PPO uses four full-batch epochs per 32-slot rollout, γ=0.99, GAE λ=0.95, clipping at 0.2, and Adam learning rates of 3×10^{-4} for the actor and 10^{-3} for the critic. Entropy regularization and gradient clipping are not used. The remaining parameters are V=1, $Z_{\max}$=10, β=0.05, and CVaR α=0.25. Federated methods aggregate at every iteration and reset the optimizer state after each broadcast. q-FFL weights clients using min–max-normalized rollout cost with exponent q=1 and a floor of 0.25; AFL applies mirror ascent, $\lambda_k \propto \lambda_k e^{0.5\,\mathrm{cost}_k}$; and clustered FL averages within equal-$\mathcal{K}$ groups. No method-specific hyperparameter tuning is applied.

d) Pre-specified metrics: The metrics are (i) the number of APs satisfying both $\hat{V}_i^{99} \le \varepsilon_{99}$ and $\hat{V}_i^{99.9} \le \varepsilon_{99.9}$, (ii) worst-AP $p_{99.9}$, and (iii) network p_{99} and $p_{99.9}$. Two-sided Mann–Whitney tests are applied at the seed level, using training seeds for learned methods and evaluation seeds for static baselines; samples are not pooled across seeds. Holm correction is applied to two 12-test families: LyMAPPO versus RR, RSSI, and SLCI over four metrics, and the independent base versus four aggregation methods over three metrics. Because learned and static methods have different statistical units, the comparison with static baselines also includes a hierarchical bootstrap over the common channel seeds $\{10, 12, 18\}$, with training and channel seeds resampled separately (Section V-B). The symbols p_{99} and $p_{99.9}$ denote the 20/40 ms deadline-violation rates $\bar{V}^{99}$ and $\bar{V}^{99.9}$ rather than latency percentiles. Training-seed indices 10 and 12 numerically coincide with two channel-seed indices. Excluding the four corresponding cells changes the full-method means by at most 0.01 feasible APs and 2×10^{-4} in network p_{99}; therefore, the cells are retained.

B. Performance at the selected operating point

Table I shows that LyMAPPO achieves an average of 15.08 feasible APs and a network p_{99} of 0.0017. The network

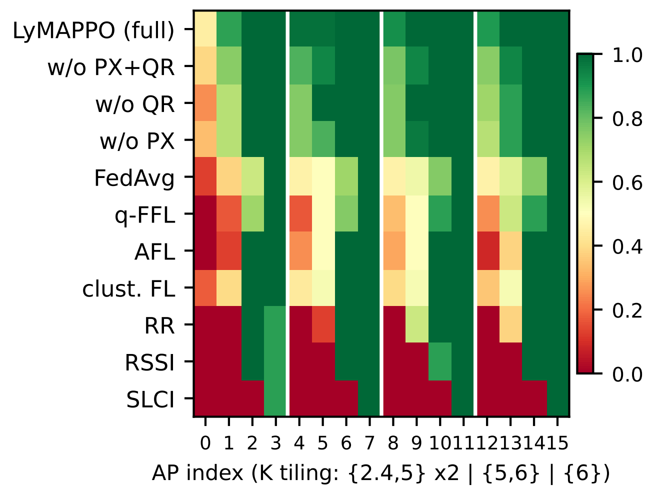

Fig. 3. Per-AP UHR feasibility rate across evaluation runs (arm $\times$ AP; feasible: $\hat{V}^{99} \le 1\%$ and $\hat{V}^{99.9} \le 0.1\%$; white rules mark the $\mathcal{K}$ tiling every four APs). All LyMAPPO failures fall in the $\mathcal{K}$=$\{2.4, 5\}$ class; the $\{5, 6\}$ and single-link $\{6\}$ classes are feasible in every run.

p_{99} is 6.1×, 9.3×, and 25× lower than for RR, RSSI, and SLCI, respectively. Differences in feasibility and network p_{99} are significant for all three baselines after Holm correction (raw $p \le .0002$, adjusted $p < .002$). A hierarchical bootstrap over common channel seeds with 20,000 resamples gives the following 95% percentile confidence intervals. The feasibility differences relative to RR, RSSI, and SLCI are +6.1 [5.1, 7.0], +7.1 [6.7, 7.4], and +11.1 [10.7, 11.4] APs. The corresponding network-p_{99} differences are -0.0066 $[-0.0105, -0.0033]$, -0.013 $[-0.021, -0.007]$, and -0.037 $[-0.040, -0.034]$. For network $p_{99.9}$, the unadjusted differences are significant for RR, RSSI, and SLCI (p=.018, .005, and .0001), but the Holm-adjusted difference is significant only for RSSI and SLCI (p=.027 and .001; RR: p=.053). Worst-AP $p_{99.9}$ differences are significant before correction (p=.028, .014, and .014) but not after correction. Delivered rates range from 4754 to 4794 pkt/s out of 4800 for all methods, so the reliability differences are not attributable to lower delivered throughput. The UHR constraints are satisfied by an average of 15.1 rather than all 16 BSSs, and the network $p_{99.9}$ of 0.0013 is slightly above the per-BSS target. The reported UHR result therefore applies per BSS to most, but not all, of the evaluated network.

a) Conservative (CI-based) feasibility: Each AP-run in the 110 s protocol contains a median of 33,000 decided packets. Under a conservative criterion, an AP is feasible only when the Wilson 95% upper confidence bound for both violation rates is below the corresponding threshold. LyMAPPO then achieves 14.34±0.95 feasible APs, compared with at most 8.50±0.93 for the static baselines. The reduction of 0.74 AP relative to the point-estimate criterion occurs in near-threshold $\{2.4, 5\}$ cells. Feasibility estimates from the 11 s and 110 s protocols differ by at most 0.4 APs where both are available. At 16-BSS density, SLCI directs multiple BSSs to the same momentarily least-loaded band, increasing synchronized contention and resulting in 3.9/16 feasible BSSs (cf. [14]). The recurrent-SAC result in [2] reports a 35% throughput-drop

TABLE I
MAIN COMPARISON AT THE SETTLED OPERATING POINT (λ=60 PKT/S/STA, SPREAD 0.6, DEADLINES 20/40 MS; 110 S EPISODES). MEAN±STD OVER TRAINING SEEDS FOR LYMAPPO (n=16) AND OVER n=8 EVALUATION CHANNEL SEEDS FOR STATIC BASELINES; RAW MWU p VS. LYMAPPO (HOLM-CORRECTED SIGNIFICANCE IN TEXT). DELIVERED RATE IS OUT OF 4800 PKT/S OFFERED.

	feasible/16 ↑	net p_{99} ↓	net $p_{99.9}$ ↓	worst-AP $p_{99.9}$ ↓	delivered pkt/s ↑
LyMAPPO (ours, full)	**15.08±0.63**	**0.0017±0.0018**	**0.0013±0.0018**	**0.0158±0.0266**	**4794±8**
RR	9.00±1.20 (p=.0001)	0.0103±0.0079 (p=.0002)	0.0040±0.0061	0.0304 (p=.028)	4783±29
RSSI	7.75±0.71 (p=.0001)	0.0158±0.0104 (p=.0001)	0.0068±0.0085	0.0489 (p=.014)	4773±41
SLCI [15]	3.88±0.35 (p=.0001)	0.0423±0.0094 (p=.0001)	0.0097±0.0076	0.0467 (p=.014)	4754±36

TABLE II
LEARNED BASELINES UNDER THE 11 S DEVELOPMENT PROTOCOL (MEAN±STD OVER n=8 TRAINING SEEDS). ALL UNADJUSTED MWU COMPARISONS WITH LYMAPPO HAVE $p\geq$.17, AND NONE IS SIGNIFICANT AFTER HOLM CORRECTION. MAPPO SHARES ONE ACTOR ACROSS 16 APS AND HAS INTER-AP WEIGHT DISTANCE 0. PPO-LAGRANGIAN USES INDEPENDENT ACTORS AND PER-AP PID-LAGRANGIAN MULTIPLIERS, WITH INTER-AP DISTANCE 9.9; IT IS DISTINCT FROM TRUST-REGION CPO [20]. LYMAPPO ACHIEVES 14.6/16 FEASIBLE APS UNDER THE 11 S PROTOCOL AND 15.1/16 UNDER THE 110 S PROTOCOL (TABLE I).

	feas./16 ↑	net p_{99} ↓	net $p_{99.9}$ ↓	worst $p_{99.9}$ ↓
LyMAPPO (ours)	**14.58±0.85**	**0.0014±0.0015**	**0.0009±0.0014**	**0.0104±0.0233**
PPO-Lagrangian	13.83±1.52	0.0026±0.0022	0.0017±0.0018	0.0230±0.0290
MAPPO	12.42±3.18	0.0075±0.0108	0.0037±0.0042	0.0383±0.0394

TABLE III
FEDERATED VARIANTS (n=8 EACH; LYMAPPO ROWS n=16) COMPARED WITH INDEPENDENT LEARNING UNDER THE SAME PROTOCOL. ALL FEDERATED VARIANTS AGGREGATE THE BASE LEARNER WITHOUT PX OR QR. THE BASE ROW PROVIDES THE INDEPENDENT COMPARISON WITH THE SAME LEARNER, AND THE FULL METHOD IS INCLUDED FOR REFERENCE. CLUSTERED FL AVERAGES WITHIN EQUAL-$\mathcal{K}$ GROUPS.

	feas./16 ↑	net p_{99} ↓	worst $p_{99.9}$ ↓
LyMAPPO (ours, full)	**15.08±0.63**	**0.0017±0.0018**	0.0158±0.0266
LyMAPPO base (indep.)	14.27±2.02	0.0026±0.0030	0.0176±0.0277
clustered FL [13]	11.17±3.58	0.0263±0.0364	0.0484±0.0566
FedAvg [10]	10.33±4.88	0.0102±0.0127	0.0153±0.0235
q-FFL [11]	9.75±3.33	0.0090±0.0130	**0.0149±0.0254**
AFL [12]	10.13±2.45	0.0185±0.0157	0.0458±0.0403

improvement over SLCI; this comparison is indirect because the present study evaluates deadline-violation rates.

Figure 3 reports the per-AP results. The single-link $\mathcal{K}$={6} APs and the $\{5, 6\}$ APs are feasible in all 47 LyMAPPO evaluation runs. All 43 infeasible AP-run pairs occur in the $\mathcal{K}$=$\{2.4, 5\}$ class. AP 0, which has the highest offered load at $\times$1.6, is feasible in 21 of 47 runs. This class has no 6 GHz link and shares the 20 MHz 2.4 GHz channel and the 5 GHz contention domain, which contains 12 BSSs. The identity of the infeasible $\{2.4, 5\}$ AP varies across training seeds. The standard deviation of the feasible-AP count is 0.63 for the full method and 2.02 for the base method; Section V-E evaluates the contribution of PX to this difference.

b) Learned baselines: We train two learned baselines in the same ns-3-in-the-loop environment. They use the same 106-dimensional state, masked per-STA link action, throughput utility, 100-iteration training budget, 67 s persistent episode, PPO hyperparameters, and checkpoint rule as LyMAPPO. No baseline-specific tuning is performed. **MAPPO** [21] uses one shared actor and the same centralized critic, with experience pooled across APs under their respective link masks. The resulting actor parameters are identical across APs, giving inter-AP weight distance 0. **PPO-Lagrangian** [22] uses 16 independent actors, with inter-AP distance 9.9, and replaces the rate-form Lyapunov dual with a per-AP PID-Lagrangian multiplier for the same UHR constraint. Each λ_i^ℓ is fixed within a rollout and updated once per iteration by a PI controller with k_P=0.25, k_I=1, and k_D=0 using the budget-normalized aggregate violation. The multiplier is not included in the observation. The rate-form Z_i is set to zero, and the constraint enters only through $r_i = Vu_i - \sum_\ell \lambda_i^\ell g_i^\ell$, avoiding duplicate cost terms. This baseline is Lagrangian PPO rather than trust-region CPO [20]. All three methods are evaluated under the 11 s protocol for the comparison in Table II. LyMAPPO has the best point estimate for each metric, but all differences have MWU $p\geq$.17 and none remains significant after correction. MAPPO has a standard deviation of 3.2 feasible APs, similar to the high variance of the parameter-averaging methods. PPO-Lagrangian preserves AP-specific policies and has a standard deviation of 1.5 feasible APs. These results are consistent with per-AP policy independence contributing more to reliability than the specific form of the constraint multiplier. The observed advantage of LyMAPPO over these learned baselines is therefore reported as a non-significant point-estimate difference under the available seed budget.

C. *Federation study: training-time benefits, held-out trade-offs*

On the training distribution, the federated and independent methods have similar behavior: all dual variables decrease, and training-window p_{99} falls approximately one order of magnitude below ε_{99} within several iterations (Fig. 5). Table III evaluates the same policies on held-out channel conditions. Relative to the independent base, the feasibility difference is significant for q-FFL (9.75±3.33, MWU p=.0013) and AFL (10.13±2.45, p=.0012), with both differences remaining significant after Holm correction (adjusted $p\leq$.015). The network-p_{99} difference for AFL is also significant after correction (p=.0036, adjusted p=.036). The unadjusted feasibility and network-p_{99} comparisons for clustered FL are p=.047 and p=.040, respectively, and do not remain significant after correction. The feasibility comparison for FedAvg is not significant (10.33±4.88, p=.071). None of the federated variants improves on the independent base for any pre-specified metric.

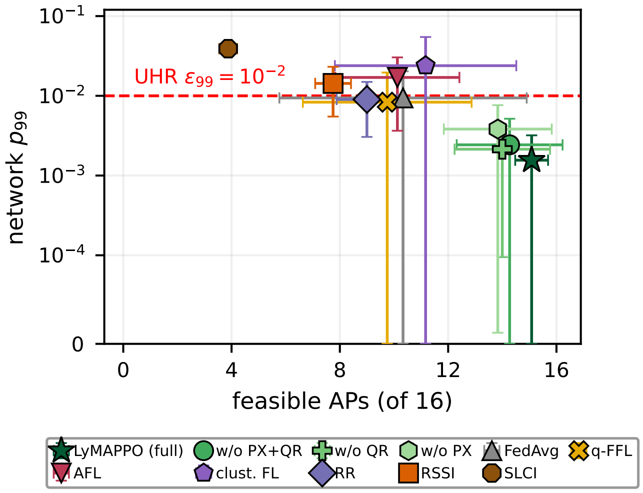

Fig. 4. Feasibility and reliability by method. Markers show means and whiskers show one standard deviation at the seed level, using training seeds for learned methods and evaluation seeds for static baselines, consistent with the tables. The horizontal axis is the number of feasible APs out of 16; the vertical axis is network p_{99} on a symmetric-log scale, with the dashed line indicating ε_{99}. Federated methods have lower mean feasibility and larger dispersion than the independent variants.

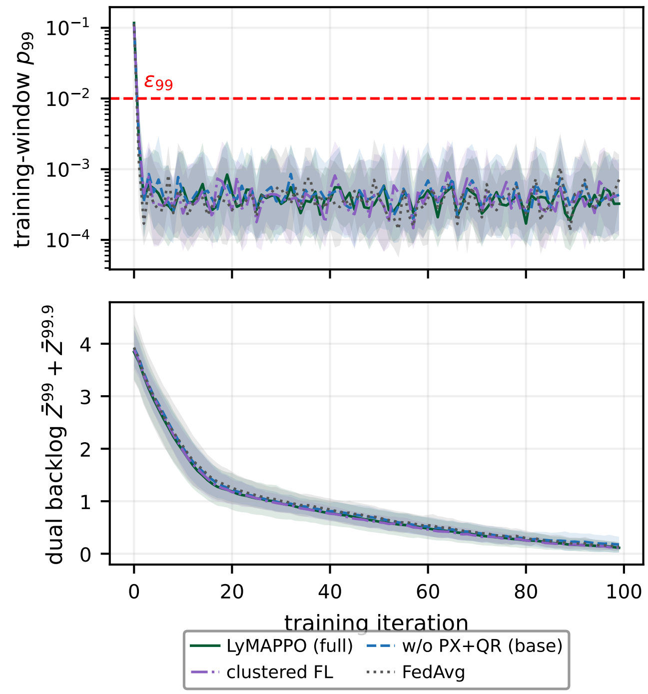

Fig. 5. Training dynamics. Lines show across-seed means, and shaded regions show one standard deviation at the seed level; line style and color identify the methods. Values of p_{99} below 10^{-4} are clipped for logarithmic display. Independent and federated methods satisfy the training constraint within several iterations and have similar dual trajectories. Their separation on held-out conditions in Table III and Fig. 4 is therefore a generalization difference rather than a training-constraint difference.

q-FFL and AFL are also not significantly different from FedAvg ($p \geq .53$), with equal or worse point estimates. Delivered rates range from 4777 to 4788 pkt/s out of 4800 across these methods, indicating that the held-out differences arise from violation rates rather than aggregate delivered throughput.

Across-seed standard deviations in the feasible-AP count are 2.5–4.9 for the federated variants, compared with 2.02 for the independent base and 0.63 for the full method (Fig. 4). In low-feasibility federated runs, all APs in the $\{2.4, 5\}$ class, and sometimes additional APs, violate at least one constraint. This can reduce the apparent worst-AP $p_{99.9}$ mean because some runs already fail for that class at the p_{99} constraint. The mean pairwise L_2 distance among the 16 independent actors is 9.95. FedAvg, q-FFL, and AFL assign the same aggregated actor to every AP, so their inter-AP distance is zero. Because weight-space distance is sensitive to hidden-unit permutations, we also compare actions. Replaying per-AP observations from one held-out run, the common FedAvg actor selects a different link from the corresponding independent actor for 40% of $\{2.4, 5\}$ decisions and 48.7% of $\{5, 6\}$ decisions. The action-distribution differences are JS ≈ 0.002 and KL ≈ 0.01. Single-link $\{6\}$ APs have no alternative link and therefore agree in all decisions. These results show that the effect of aggregation is behavioral as well as parametric. The aggregated actor is at approximately equal distance, about 7.25, from the independent actors of each band class. Thus, aggregation reduces AP-specific specialization across all classes; the $\{2.4, 5\}$ class becomes infeasible first because it has the least reliability margin. Fairness-based client weighting modifies aggregation weights but does not preserve separate client policies. Clustered FL retains cross-class differences by averaging only within equal-$\mathcal{K}$ groups, giving an inter-AP distance of 2.97 and the highest feasibility point estimate among the federated variants (11.17±3.58). It nevertheless removes within-class policy differences and does not match the independent base on held-out seeds. PX provides cross-AP congestion information through one scalar per AP while retaining separate policies (Section IV).

D. Robustness across load

Figure 6 evaluates loads below and above the operating point in Table I. At λ=40 and 80, LyMAPPO achieves 15.50±0.69 and 11.38±1.47 feasible APs, respectively. Across the same load range, RR changes from 13.33±0.58 to 8.00±0.00, RSSI from 12.67±1.15 to 8.00±0.00, and SLCI from 12.00±3.00 to 3.67±0.58. LyMAPPO's network-p_{99} is 8–28× lower than the static baselines across the sweep. All methods improve at lower load without changing their relative ordering. At λ=80, RR and RSSI satisfy both constraints only for the $\{5, 6\}$ and $\{6\}$ classes, whereas LyMAPPO retains an average of 11.4 feasible APs.

E. Ablations

a) Component ablation (PX, QR): With n=16 seeds per method, removing both PX and QR changes feasibility from 15.08±0.63 to 14.27±2.02, network p_{99} from 0.0017 to 0.0026, and worst-AP $p_{99.9}$ from 0.0158 to 0.0176. The mean feasibility difference is not significant (MWU p=.39).

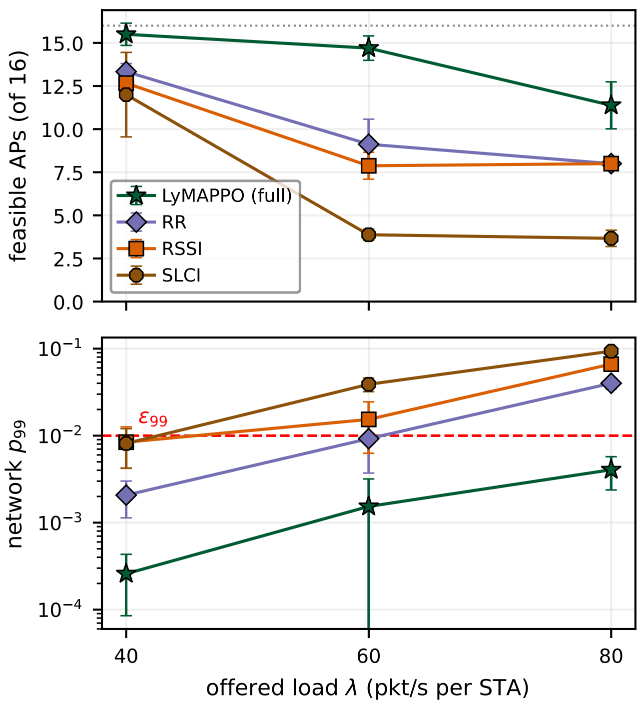

Fig. 6. Load sweep for $\lambda \in \{40, 60, 80\}$ pkt/s/STA under the 11 s development protocol. LyMAPPO is retrained at each load, whereas static baselines require evaluation only. The top panel shows feasible APs and the bottom panel shows network p_{99} on a logarithmic scale, with the dashed line indicating ε_{99}. At λ=80, the static link-selection methods satisfy both constraints only for the eight APs outside the $\{2.4, 5\}$ class.

Bootstrap estimates give a full-minus-base feasibility difference of +0.81 APs with 95% CI $[-0.08, +1.90]$; the network-p_{99} interval also includes zero. In contrast, the difference in feasible-AP standard deviation is significant: the base-minus-full estimate is +1.39 with 95% CI $[+0.23, +2.39]$. Thus, the data support a reduction in across-seed dispersion but not a mean improvement over the base method. The individual components achieve 14.00±1.89 feasible APs with PX only and 13.83±2.13 with QR only (n=8), both below the base point estimate. One possible post-hoc explanation is that PX changes the reward distribution while QR changes the advantage baseline, so the two components may interact. This interpretation remains a hypothesis. The statistically established comparison is the full method's feasibility and network-p_{99} difference relative to the three static baselines after Holm correction (Table I).

b) Hyperparameter sensitivity (PX, QR): Table IV varies one parameter at a time around the selected configuration. The β=0 and selected configurations reuse the no-PX and full-method results. Mean feasibility increases and network p_{99} decreases from β=0 to β=0.05, with little additional change over $\beta \in [0.025, 0.1]$. Pairwise endpoint comparisons are underpowered with n=8–16 and are not used as significance claims. For QR, neither the tail fraction α nor the quantile count K produces differences beyond the observed seed variation. One incomplete evaluation cell is excluded as documented.

TABLE IV

SENSITIVITY TO PX AND QR HYPERPARAMETERS UNDER THE 11 S DEVELOPMENT PROTOCOL (SECTION V-A). VALUES ARE MEAN±STD OVER TRAINING SEEDS, WITH n=16 AT THE SELECTED CONFIGURATION AND n=8 ELSEWHERE. PERFORMANCE VARIES LITTLE OVER $\beta \in [0.025, 0.1]$ AND OVER THE EVALUATED QR SETTINGS.

	feasible/16 ↑	net p_{99} ↓
PX weight β (QR fixed: K=8, α=0.25)		
0 (w/o PX)	13.46±2.15	0.0041±0.0042
0.01	13.79±1.66	0.0028±0.0024
0.025	13.96±1.40	**0.0015±0.0015**
0.05 (chosen)	**14.70±0.73**	0.0016±0.0018
0.1	14.42±1.34	**0.0015±0.0015**
CVaR fraction α (β=0.05, K=8)		
0.1	14.15±1.70	0.0020±0.0026
0.25 (chosen)	**14.70±0.73**	**0.0016±0.0018**
0.5	14.21±0.96	0.0021±0.0018
Quantile count K (β=0.05, α=0.25)		
8 (chosen)	**14.70±0.73**	**0.0016±0.0018**
16	14.46±0.99	0.0017±0.0016
32	14.63±1.01	0.0017±0.0016

c) Removing the dual layer (no-Z): For matched seeds $s \in \{0..7\}$, the no-Z method and the base learner achieve 14.46±1.21 and 14.24±1.49 feasible APs, respectively. Their worst-AP $p_{99.9}$ values are 0.0159 and 0.0195, and network p_{99} is also not significantly different (MWU $p \geq .62$). One no-Z training run did not complete and is excluded. These results do not establish a direct performance contribution from the dual layer at this operating point. The drop-inclusive service process already includes deadline violations in the throughput objective and may provide a sufficient learning signal in this setting. The dual layer nevertheless provides two functions: virtual-queue stability can be monitored at run time for Slater-feasible APs (Section IV-A), and Z_i supplies the congestion price used by PX (Section IV). When the constraint is slack, Z_i approaches zero and the associated penalty vanishes.

d) Environment calibration probe (link-2 width): With an 80 MHz 6 GHz link, the Markov channel produces 180 link-2 drops per episode, compared with 13 under a static channel. Reducing the link width to 40 MHz lowers the drop count to 12 under the same Markov process. Because this effect applies to all policies, the evaluation fixes the 6 GHz bandwidth at 40 MHz rather than selecting it separately for each method.

F. Limitations

Band-class limitation. All observed infeasible AP-run pairs occur in the $\mathcal{K}$=$\{2.4, 5\}$ class (Fig. 3). At this operating point, feasibility is limited by spectrum access: these APs have no 6 GHz link, share a 12-BSS 5 GHz contention domain, and use a 20 MHz 2.4 GHz link. Single-link $\{6\}$ APs are feasible in all evaluated runs. Results from the 80 MHz calibration setting do not apply to the selected 40 MHz protocol.

Seed dependence. The identity of the infeasible $\{2.4, 5\}$ AP varies with the training seed. The standard deviation of the feasible-AP count is 0.63 for the full method and 2.02 for the base method (n=16 each). We report mean±std over all completed seeds without selecting results by seed.

Simulator scope. MLO is emulated using per-link 802.11ax devices rather than a native 802.11be/bn MLO implementation, and rate adaptation is idealized. The main results use one operating point and one held-out channel-seed family. Channel seed 11 is excluded a priori because it triggers a pre-existing ns-3 PHY assertion; four evaluation cells with the same assertion are also excluded and documented.

Learned baselines. The learned comparisons include shared-actor MAPPO and independent PPO-Lagrangian trained in the same environment (Table II, Section V-B), in addition to four parameter-aggregation methods (Section V-C). LyMAPPO has the best point estimate for each metric, but its differences from MAPPO and PPO-Lagrangian are not significant with eight seeds. A larger seed set is required to estimate these differences more precisely. The recurrent-SAC comparison is indirect because it uses published throughput-drop results rather than a reimplementation in the present environment.

Guarantee gap. Theorem 1 bounds long-run *average* violation rates; the per-sample tail quantiles we report are empirical (Remark 1).

VI. Conclusion

This paper formulates per-BSS UHR in dense multi-band Wi-Fi as a constrained decentralized learning problem. LyMAPPO combines independent per-AP PPO, rate-form Lyapunov dual variables, one-scalar cross-BSS congestion signaling, and a lower-tail quantile baseline. In ns-3-in-the-loop evaluation, it satisfies both UHR constraints for an average of 15.1 of 16 BSSs and achieves a network p_{99} of 0.0017. Feasibility and network-p_{99} differences relative to RR, RSSI, and SLCI remain significant after multiple-comparison correction, whereas the $p_{99.9}$ comparisons are weaker. Four federated aggregation methods match the independent learner on the training distribution but have lower held-out reliability; action replay associates this difference with reduced per-AP specialization after aggregation. PX communicates congestion state while retaining separate actor parameters. The theoretical bounds include the clipping deficit and are conditional on Slater feasibility and C_{RL}. Future work will consider native 802.11be/bn MLO, multi-AP coordination actions, larger seed sets, and load–topology sweeps.

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